

YAKUSHIMA AP, Japan

WMO: 478360

Lat: 30.386N

Lon: 130.659E

Elev: 124

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.7	44.5	28.7	22.8	48.9	30.8	25.1	49.2	34.0	50.9	31.7	50.3	15.0	290	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	88.9	79.4	87.6	79.1	86.5	78.7	81.2	85.7	80.5	85.2	79.7	84.5	9.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	156.6	83.9	79.1	151.9	83.5	78.3	147.7	83.1	45.1	86.0	44.2	85.2	43.5	84.7	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.4	26.8	23.6	39.0	91.6	2.0	1.7	37.6	92.9	36.4	93.8	35.3	94.8	33.8	96.0		
(5)			35.5	82.5	1.5	1.6	34.4	83.6	33.6	84.6	32.7	85.5	31.7	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.4	52.9	54.2	58.0	63.8	69.8	74.8	80.9	81.8	78.5	72.3	64.6	56.9	(6)
(7)		DBStd	10.80	4.90	5.46	4.99	4.24	3.20	3.80	2.30	1.84	2.92	3.86	4.76	5.27	(7)
(8)		HDD50	50	24	16	4	0	0	0	0	0	0	0	0	6	(8)
(9)		HDD65	1298	375	304	223	70	3	0	0	0	0	2	64	257	(9)
(10)		CDD50	6414	114	133	251	415	614	743	956	985	855	691	439	218	(10)
(11)		CDD65	2190	1	2	5	36	152	293	491	520	405	228	53	4	(11)
(12)		CDH74	16988	0	0	5	30	357	1594	5079	5643	3403	826	50	1	(12)
(13)		CDH80	4422	0	0	0	0	13	275	1565	1819	696	52	2	0	(13)
(14)	Wind	WSAvg	10.4	13.2	12.9	12.3	10.8	9.2	7.3	7.2	8.3	9.1	10.6	11.0	12.7	(14)
(15)	Precipitation	PrecAvg	168.0	10.6	10.2	13.6	15.0	16.2	31.3	12.9	9.4	16.0	11.5	11.5	9.9	(15)
(16)		PrecMax	231.5	22.6	26.0	28.8	28.0	32.0	49.3	35.4	21.7	44.9	29.8	22.5	23.7	(16)
(17)		PrecMin	82.7	0.7	3.7	3.9	6.3	4.4	12.8	0.4	1.5	3.1	1.1	4.8	2.0	(17)
(18)		PrecStd	25.9	4.6	4.8	5.5	6.0	7.4	10.0	9.3	5.8	10.4	7.5	5.1	4.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.0	71.0	73.5	77.1	81.7	88.4	90.7	90.8	88.2	83.7	78.3	71.8	(19)
MCWB			62.7	64.8	64.4	68.4	72.0	78.0	78.4	79.7	79.2	75.8	71.8	64.8	(20)	
2%		DB	65.7	67.9	70.1	74.4	79.0	85.6	88.8	89.1	86.4	81.4	75.2	68.9	(21)	
		MCWB	59.8	63.0	62.8	67.4	69.4	77.4	79.2	79.7	78.3	74.8	68.5	62.6	(22)	
5%		DB	63.2	65.4	68.0	72.6	77.4	83.3	87.6	87.8	85.0	79.7	73.5	67.0	(23)	
		MCWB	57.8	60.4	61.7	66.0	69.1	76.8	79.0	79.5	77.5	73.2	66.9	61.2	(24)	
10%		DB	60.9	62.9	66.1	71.0	76.0	80.8	86.4	86.7	83.8	78.1	71.8	65.1	(25)	
		MCWB	55.3	58.0	60.1	64.5	68.6	75.7	78.7	79.0	76.8	71.7	65.6	59.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	66.8	68.2	72.6	75.9	79.9	82.3	82.5	81.2	78.8	73.4	67.3	(27)
MCD B			67.5	69.5	70.6	74.5	78.9	84.7	86.5	86.6	85.5	81.0	76.5	69.3	(28)	
2%		WB	61.3	63.8	65.7	70.5	74.2	78.5	81.0	81.4	79.9	76.7	70.7	64.6	(29)	
		MCD B	64.5	66.8	68.3	72.7	76.8	82.9	85.7	85.9	84.0	79.8	73.7	67.7	(30)	
5%		WB	58.9	61.3	63.5	68.4	72.7	77.5	80.2	80.7	79.0	75.1	68.8	62.6	(31)	
		MCD B	62.6	64.6	66.5	71.0	75.1	81.6	85.1	85.4	83.0	78.5	72.0	66.1	(32)	
10%		WB	56.0	58.9	61.4	66.4	71.1	76.5	79.5	80.0	78.2	73.5	67.1	60.2	(33)	
		MCD B	60.2	62.2	64.7	69.4	73.8	80.2	84.5	84.8	82.2	77.0	70.7	64.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.9	9.6	10.6	11.3	11.1	9.0	10.2	10.0	9.6	9.1	9.4	9.0	(35)
MCDBR			13.0	12.8	13.4	13.3	13.2	11.6	12.2	11.5	10.9	10.5	10.5	11.0	(36)	
MCWBR			10.8	10.9	10.6	9.7	9.2	6.4	6.7	6.2	6.3	6.6	7.7	8.8	(37)	
5% WB		MCD BR	11.7	11.5	11.0	10.1	9.6	10.0	10.8	10.1	9.5	8.4	8.8	9.0	(38)	
		MCWBR	11.2	11.3	11.0	9.4	8.4	6.1	6.6	5.8	6.0	6.1	8.0	8.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.444	0.476	0.528	0.540	0.531	0.523	0.500	0.494	0.486	0.474	0.444	0.437	(40)
taud		2.094	2.039	1.911	1.939	2.034	2.149	2.242	2.259	2.239	2.185	2.193	2.113	(41)		
Ebn at Noon		241	246	242	245	247	248	253	254	252	244	240	236	(42)		
Edh at Noon		43	49	60	60	55	49	44	43	43	42	38	40	(43)		
(44)	All-Sky Solar Radiation	RadAvg		677	878	1195	1441	1524	1323	1863	1830	1477	1157	819	663	(44)
(45)		RadStd		73	108	90	137	170	158	174	109	149	128	97	66	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page